CS3009 - Software Engineering

Project Management

Lecture # 13,14,15,16

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TODAY'S OUTLINE

- **Project management**
- **Introduction(Project Planning)**
- **What is Work Breakdown Structure(WBS)?**
- **WBS Concepts**
- **WBS Goals**
- **When should we develop WBS?**
- **Why to develop WBS?**
- **How to Develop WBS**
- **WBS levels, pitfalls, guidelines, check list**
- **Wideband Delphi technique**
- **Revision**

PROJECT MANAGEMENT

Planning, Estimating, Scheduling

- What's the difference?
- Plan: Identify activities. No specific start and end dates.
- Estimating: Determining the size & duration of activities.
- Schedule: Adds specific start and end dates, relationships, and resources.

WHY (CONT…)

- You need to decompose your project into manageable chunks
- ALL projects need this step
- Divide & Conquer
- Two main causes of project failure
  - Forgetting something critical
  - Ballpark estimates become targets

INTRODUCTION

- **Dividing complex projects to simpler and manageable tasks is the process identified as Work Breakdown Structure (WBS).**

- **Usually, the project managers use this method for simplifying the project execution. In WBS, much larger tasks are broken-down to manageable chunks of work. These chunks can be easily supervised and estimated.**
- **Work Package:** **A group of related tasks that are defined at the same level within a work breakdown structure.**

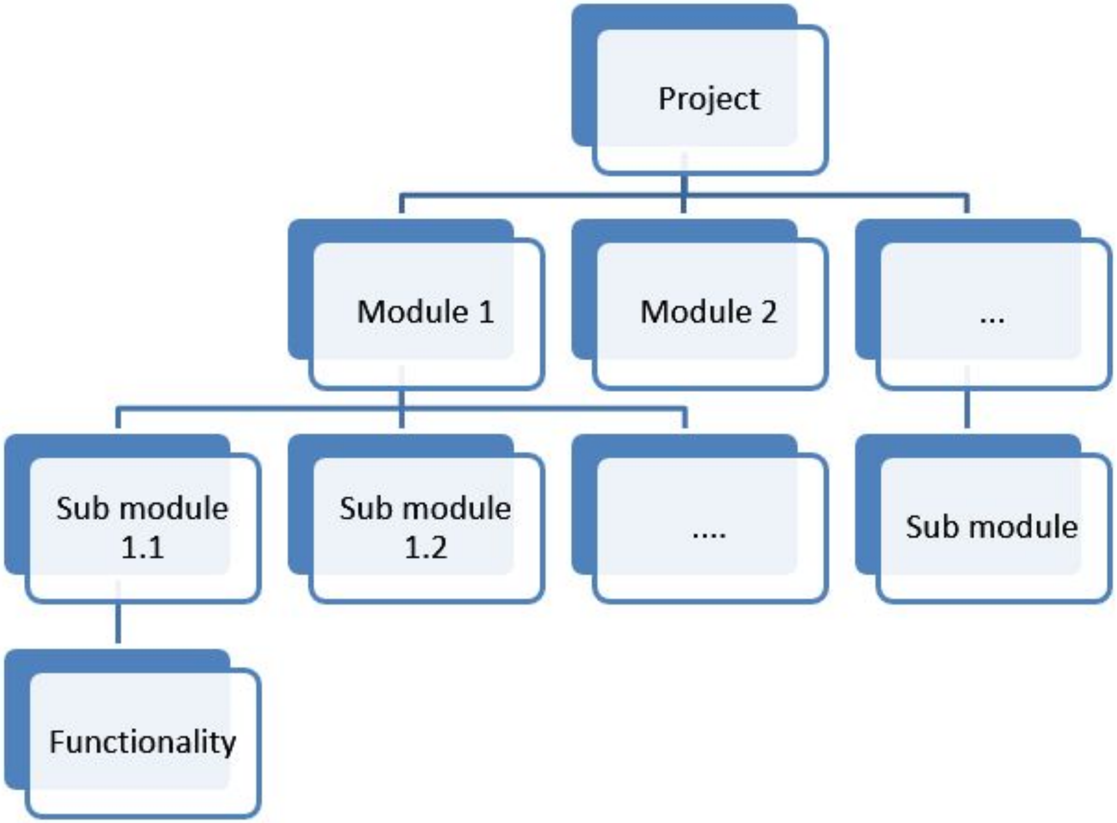
WHAT IS WBS?

- **In project management and systems engineering, WBS is a deliverable oriented decomposition of a project into smaller components.**

- **A work breakdown structure element may be a product, data, a service, or any combination. A WBS also provides the necessary framework for detailed cost estimating and control along with providing guidance for schedule development and control**

THE WORK BREAKDOWN STRUCTURE

- Used as a basis to produce the subsidiary plans of the Project Management Plan.
- The WBS is a deliverable-oriented hierarchy of decomposed project components that organises and defines the total scope of the project. The WBS is a representation of the detailed project scope statement that specifies the work to be accomplished by the project.
- The elements comprising the WBS assist the stakeholders in viewing the end product of the project.
- The work at the lowest-level WBS component is estimated, scheduled, and tracked.


Project
Module 1
Module 2
...
Sub module 1.1
Sub module 1.2
....
Sub module
Functionality

WBS CONCEPTS

- **Select a suitable category (work, product or other relevant structure)**
- **is not constrained by sequence**
- **final box is a product or deliverable which is measurable and definable**
- **lowest level indicate work packages, which can be used for estimates, schedule monitoring and control**
- **Entire project team should be involved**

GOALS FOR WBS

- **Giving visibility to important work efforts.**

- **Giving visibility to risky work efforts.**

- **Illustrate the correlation between the activities and deliverables.**

- **Show clear ownership by task leaders.**
- **Breaking down a project into tasks allows effective delegation**
- **Individual tasks allow people to focus**
- **Without a WBS, things will be forgotten, started late, or allocated to several people.**

WHEN SHOULD WE DEVELOP WBS?

- **Once the project *Scope* is agreed (finalized) then before starting the project we need to *plan* various *components* (activities) of software development project.**

WBS HELPS MANAGER

- **Facilitates evaluation of cost, time, and technical performance of the organization on a project.**
- **Provides management with information appropriate to each organizational level.**
- **Helps organization to project responsibilities to organizational units and individuals**
- **Helps manager plan, schedule, and budget.**
- **Defines communication channels and assists in coordinating the various project elements.**

WBS GUIDELINES

- **Accurate and readable project organization.**
- **Accurate assignment of responsibilities to the project team.**
- **Indicates the project milestones and control points.**
- **Helps to estimate the cost, time, and risk.**
- **Illustrate the project scope, so the stakeholders can have a better understanding of the same.**

GUIDELINES - CONTINUED

- Include three types of project work
  - Product
    - **Specifically assigned to a physical product as a unique deliverable**
    - **This subset is sometimes referred to as the product breakdown structure**
  - Integration
    - **When products are brought together as a unit**
    - **Can be at any level**
  - Support
    - **Level of Effort, Administration, Expenses, Improvement Practices, Contractor Management**

STEPS TO BUILD A WBS

- **Begin with the Charter, focusing on Objectives and Deliverables**
- **Break the main product(s) down into sub-products**
- **Set the structure to match how you'll manage the project**
- **Lowest level not too detailed, not too large**
- **Is there a need for Integration?**
- **Identify support activities**
- **Check for completeness - is all the effort included?**
- **Develop a coding structure if needed**
- **Assign work package managers**

WBS FORMATS

- 2 Formats
- Outline (Indented Format)
- Graphical Tree (Organizational Chart)

DISPLAYING THE WBS

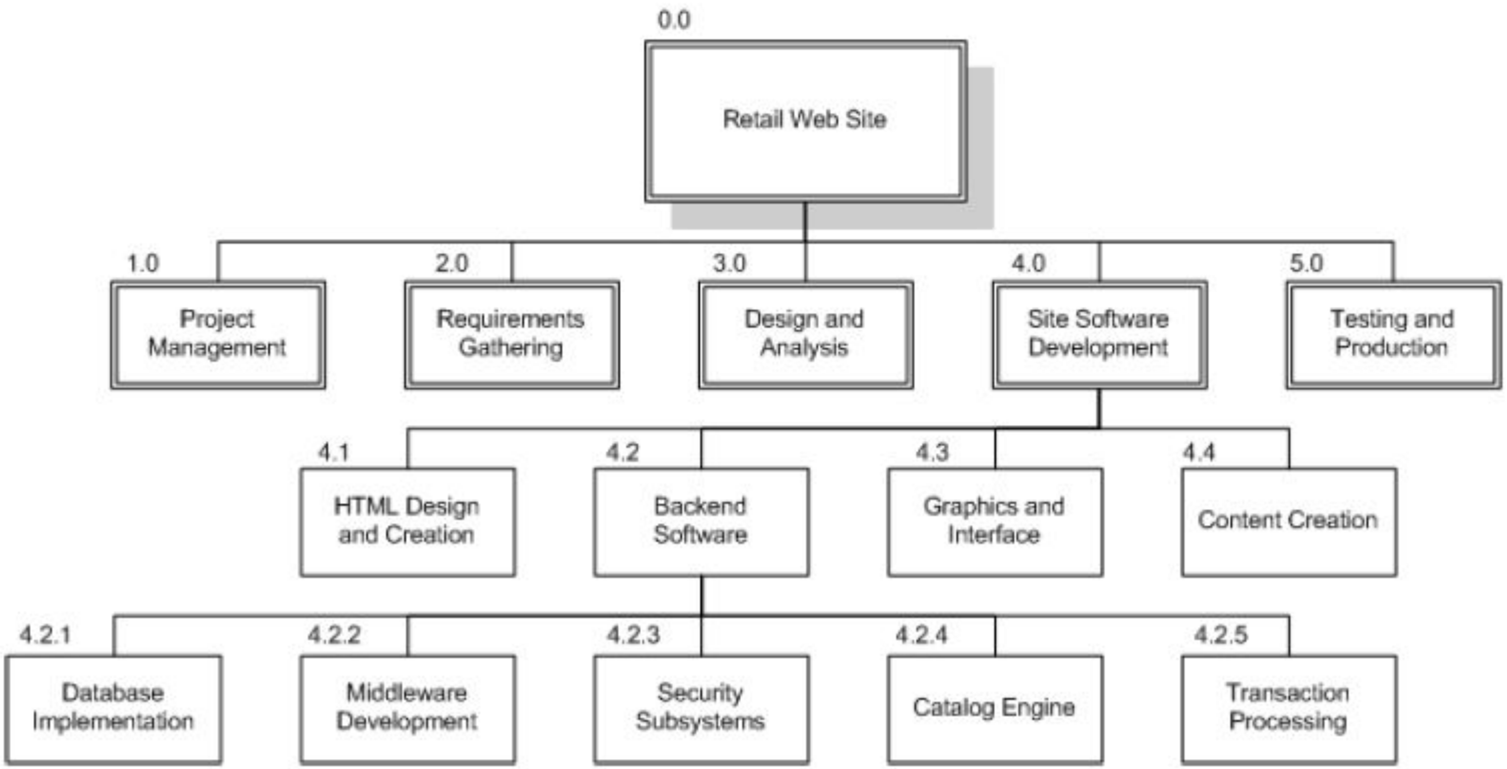
EXAMPLE OF OUTLINED WBS.

Project Name			
	Task 1		
		Subtask 1.1	
			Work Package 1.1.1
			Work Package 1.1.2
		Subtask 1.2	
			Workpackage 1.2.1
			Workpackage 1.2.2
	Task 2		
		Subtask 2.1	
			Workpackage 2.1.1
			Workpackage 2.1.2

DISPLAYING THE WBS
EXAMPLE OF OUTLINED WBS.

- 0.0 Retail Web Site
- 1.0 Project Management
- 2.0 Requirements Gathering
- 3.0 Analysis & Design
- 4.0 Site Software Development
  - 4.1 HTML Design and Creation
  - 4.2 Backend Software
    - 4.2.1 Database Implementation
    - 4.2.2 Middleware Development
    - 4.2.3 Security Subsystems
    - 4.2.4 Catalog Engine
    - 4.2.5 Transaction Processing
  - 4.3 Graphics and Interface
  - 4.4 Content Creation
- 5.0 Testing and Production

DISPLAYING THE WBS
EXAMPLE OF CHART WBS.



PITFALLS

There are common pitfalls to creating a WBS. If you can keep these few possible, you and your team will be much more successful at creating a useful and accurate Work Breakdown Structure.

- **Level of Work Package Detail**
- **Deliverables Not Activities or Tasks**
- **WBS is not a Plan or Schedule**
- **WBS Updates Require Change Control**
- **WBS is not an Organizational Hierarchy**

THE DICTIONARY OF THE WBS

- **A WBS dictionary is a <u>companion document</u> to the WBS that describes each WBS element. For each WBS element, the WBS dictionary includes a <u>statement of work, a list of associated activities, and a list of milestones</u>.**
- **Other information can include the responsible organisation, start and end dates, resources required.**

WBS LEVELS

- Contract WBS (CWBS)
  - First 2 or 3 levels
  - High-level tracking
  - For potential discussion with clients
- Project WBS (PWBS)
  - Defined by PM and team members
  - Tasks tied to deliverables
  - Lowest level tracking

A FULL WBS STRUCTURE

- Up to six levels (3-6 usually) such as

Managerial levels
1 Total Program
2 Project
3 Task

Technical levels
4 Subtask
5 Work Package
6 Level of Effort

- Upper 3 can be used by customer for reporting
- Different level can be applied to different uses
  - Ex: Level 1: authorizations; 2: budgets; 3: schedules

WBS GUIDELINES

- Should be easy to understand
- Some companies have corporate standards for these schemes
- Some top-level items, like Project Management, are in WBS for each project
  - Others vary by project
- What often hurts most is ***what's missing***
- Break down until you can generate accurate time and cost estimates
- Ensure each element corresponds to a deliverable

WBS GUIDELINES

- How detailed should it be?
  - **Not as detailed** as the final project plan
  - Each level should have no more than 7 items
  - It can evolve over time
- What tool should you use?
  - Excel, Word, Project
  - Org chart diagramming tool (Visio, etc)
  - Specialized commercial apps
- Re-use a “template” if you have one

WBS CHECK LIST

- **Does it define 100% of the work that will be produced by the project?**
- **Does it use a coding structure so that each element has a unique ID that shows its place in the hierarchy e.g. 1.1, 1.2, 1.1.1, 1.1.2?**
- **Will project stakeholders be able to understand the project scope from the WBS?**
- **Does each level represent 100% of the work required to deliver the parent level?**
- **Is the decomposition sufficient that the tasks required to deliver each work package can easily be identified?**
- **Is it in the format that gives a clear graphical, textual or tabular breakdown of the project scope?**

WBS CHECK LIST

- **Does it have a least two levels with at least one level of decomposition?**
- **Was it created by those who will be performing the work?**
- **Is it being regularly updated as project changes are approved?**
- **Can you identify one person who is accountable for each work package?**
- **Can you clearly define the acceptance criteria for each work package?**
- **Does it allow you to estimate costs accurately?**
- **Does the WBS have logical summary elements that can be used in tracking progress and performance?**

EXAMPLE WBS

- **Redecorate Room**
  - **Prepare materials**
    - **Buy paint**
    - **Buy a ladder**
    - **Buy brushes/rollers**
    - **Buy wallpaper remover**
  - **Prepare room**
    - **Remove old wallpaper**
    - **Remove detachable decorations**
    - **Cover floor with old newspapers**
    - **Cover electrical outlets/switches with tape**
    - **Cover furniture with sheets**
  - **Paint the room**
  - **Clean up the room**
    - **Dispose or store left over paint**
    - **Clean brushes/rollers**
    - **Dispose of old newspapers**
    - **Remove covers**

PROJECT ESTIMATION

WHAT IS ESTIMATION?

- The project manager must set expectations about the time required to complete the software among the stakeholders, the team, and the organization's management.
- expectations should be realistic
- If those expectations are not realistic from the beginning of the project, the stakeholders will not trust the team or the project manager.

PREREQUISITES

- Vision Statement
- Scope
- Requirements

ELEMENTS OF A SOUND ESTIMATE

- To generate a sound estimate, a project manager must have:
  - A work breakdown structure (WBS), or a list of tasks which, if completed, will produce the final product
  - An effort estimate for each task
  - A list of assumptions which were necessary for making the estimate
  - Discussion and Consensus among the project team that the estimate is accurate

WIDEBAND DELPHI

- *Wideband Delphi* is a process that a team can use to generate an estimate
  - The project manager chooses an estimation team, and gains consensus among that team on the results
  - Wideband Delphi is a *repeatable* estimation process because it consists of a straightforward set of steps that can be performed the same way each time

WIDEBAND DELPHI ROLES

Estimation Team: PM chooses an estimation team that include reps from all project areas (managers, developers, architect, QA, writers, etc).

- every team member should have stake in plan
- should understand Delphi Process

Moderator: someone who understands Delphi Process *but has no stake in the results*

Observers: selected stakeholders or users.

- encourages trust in the estimation process
- sense of ownership in the results

ENTRY CRITERIA

1. Vision and Scope documents have been agreed on by stakeholders
2. Kick-off meeting has been scheduled
3. Estimation meeting has been scheduled (1-2 hrs)
4. Moderator chosen (*not* the PM)
5. Agreement on the goal of the estimation session

THE WIDEBAND DELPHI PROCESS

- Step 1: Choose the team
  - The project manager selects the estimation team and a moderator. The team should consist of 3 to 7 project team members.
    - The moderator should be familiar with the Delphi process, but should not have a stake in the outcome of the session if possible.
    - If possible, the project manager should not be the moderator because he should ideally be part of the estimation team.

THE WIDEBAND DELPHI PROCESS

- Step 2: Kickoff Meeting
  - The project manager must make sure that each team member understands the Delphi process, has read the vision and scope document and any other documentation, and is familiar with the project background and needs.
  - The team brainstorms and writes down assumptions.
  - The team generates a WBS with 10-20 tasks.
  - The team agrees on a unit of estimation.

INDIVIDUAL PREPARATION

Step 3: Individual Preparation

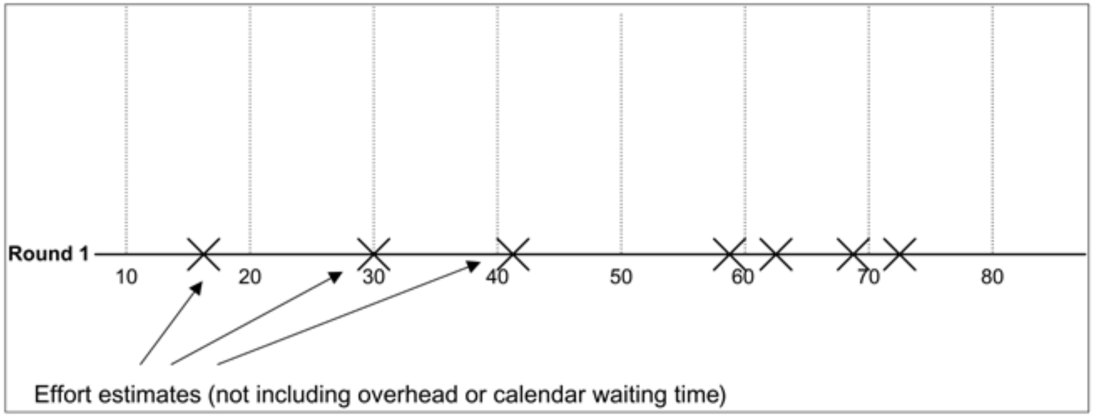
- each member generates his own initial estimates for all tasks in the WBS
  - identify subtasks may help clarify an estimate
- for each task, the team member writes:
  1. estimate of effort required to complete task
  2. any assumptions needed to make the estimate
- write down any new tasks the were missed during brainstorming session

THE WIDEBAND DELPHI PROCESS

- Step 4: Estimation Session
  - During the estimation session, the team comes to a consensus on the effort required for each task in the WBS.
  - Each team member fills out an estimation form which contains his estimates.
  - The rest of the estimation session is divided into rounds during which each estimation team member revises her estimates based on a group discussion. Individual numbers are not discussed.

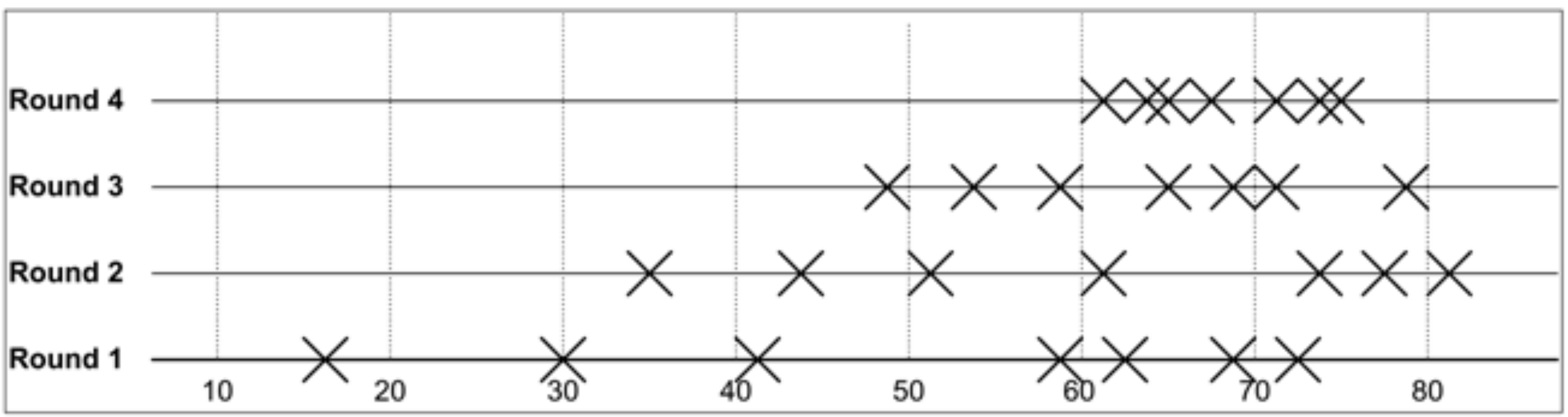
THE WIDEBAND DELPHI PROCESS

- Step 4: Estimation Session (continued)
  - The moderator collects the estimation forms and plots the sum of the effort from each form on a line:



THE WIDEBAND DELPHI PROCESS

- Step 4: Estimation Session (continued)
  - The team resolves any issues or disagreements that are brought up.
    - Individual estimate times are not discussed. These disagreements are usually about the tasks themselves. Disagreements are often resolved by adding assumptions.
  - The estimators all revise their individual estimates. The moderator updates the plot with the new total:



WIDEBAND DELPHI ESTIMATION

Step 4: Estimation Session (loop)

- moderator leads the team through several rounds of estimates to gain consensus on estimates.

Exit Criteria

- The estimation session continues until the estimates converge or the team is unwilling to revise estimates.
- Process also ends if time limit (2 hours) has elapsed.

THE WIDEBAND DELPHI PROCESS

- Step 5: Assemble Tasks
  - The project manager works with the team to collect the estimates from the team members at the end of the meeting and compiles the final task list, estimates and assumptions.
- Step 6: Review Results
  - The project manager reviews the final task list with the estimation team.

OTHER ESTIMATION TECHNIQUES

- PROBE, or Proxy Based Estimating
- COCOMO II
- The Planning Game

That is all